

ATTACHMENT A1

ARCHITECTURE & ENGINEERING DESIGN SCOPE OF SERVICES **International New Construction Temple Project**

2025 v1

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i. DESIGN DEFINITIONS:

The purpose of this section is to provide design definitions for commonly referenced terms. These terms may or may not be part of the contractual agreement, however they will likely be used in the course of project delivery and are important for consultants to understand.

Arrival Center: Patron support facility located on some temple sites that provides a space for temple patrons to shower, dress, prepare food and wait while attending the temple. It does not provide overnight patron accommodation. Arrival Centers may include a Distribution Center or Family History Center.

Ancillary Building: Refers to multiple types of patron support facilities to include Arrival and Patron Housing Buildings.

As-builts: Conformed set of drawings and specifications, updated regularly throughout construction to reflect the actual as-built conditions, locations of utilities and other actual conditions on site, for future Owner reference.

ATFM: Area Temple Facilities Manager, responsible for a global region.

BOD: Basis of Design document, milestone establishing proposed and project systems across all buildings and all major finishes and systems included in the project. Once the BOD is prepared by design consultants, it is reviewed with the Owner Team (including engineering) to make selections. Final material and system approvals are updated in the document which then serves as a guide for the remainder of the document development process.

Conformance Set: The 100% complete design documents (drawings and specification) combined with any edits, additions or changes made during the bidding and buy-out process. The conformed set is considered the official construction set of documents.

CSD: Construction Services Division, the part of the SPD group tasked with design and construction.

DCS: Design and Construction Services (DCS), the division of SPD responsible for the design and construction of physical facilities.

Deliverable A: Schematic design deliverable documenting all design decisions and elements depicted in project renderings, in an architectural format for further design development.

Deliverable B: All project drawings and specifications submitted at a 50% complete milestone review.

Deliverable C: All project drawings and specifications submitted at a 100% complete milestone review.

Deliverable D: All post-project deliverables including as-builts and record drawings, specifications, samples, project books and other completion requirements.

Department Review: Also known as PBO review, the second committee in the four-committee approval process.

Design Consultant: The architectural firm contractually responsible for the deliverables included in each phase of work included in the contract. The design consultant may include in-house services such as Engineering or Interior Design. Alternatively, the design consultant may have sub-consultant agreements for scopes of work performed by others, that they are responsible for overseeing and coordinating.

Design Manager: Special Projects Department (Owner) DM, manager of architecture for the project. Member of Owner Project Team.

Distribution Center: Patron support facility on temple sites to provide a small retail space where users can purchase items needed for temple attendance. Distribution Centers may be included in the program of an Arrival or Patron Housing building or freestanding, located on a separate parcel, depending on local tax laws.

Interior Design Consultant: Interior Designer associated with the design consultant architectural firm or a separate sub-consultant entity, responsible for the project. Responsible for interior design deliverables included in each phase of work included in the contract.

Interior Design Manager: Special Projects Department (Owner) IDM, manager of interior design for the project. Member of Owner Project Team.

Family History Center: Secondary facility on temple sites, usually part of the Arrival or Patron Housing program. This space is used to provide computer and printing stations for patrons to work on genealogical research for their families in preparation for temple attendance.

Fohlio: Specification database to manage, track and document all furniture, fixtures, equipment and finishes in a project. Budgets and progress reports are also generated from this system. End of project O&M manuals are published through Fohlio.

Leadership Approval: Indicates the monthly approval cycle represented by Project Review, Presiding Bishopric Review (PBO), Temple and Family History Executive Committee (TFHEC) and the Temple Sites Committee (TSC). A project is considered approved to move forward with design and construction when all committees have reviewed and approved the project. The Owner Project Team (PM, DM, IDM, & PA) will be the sole point of contact for the design consultant and will provide periodic approval updates.

LOD: Level of Development definitions for BIM modeling standards.

Owner Project Team: Assigned project team Owner representation which includes a PM, DM, IDM and PA. All members of the Owner team will be copied on all project correspondence. Owner Team reports to a Regional Director.

Owner Support Team: Owner representatives for specialty scope to include landscape, structural, mechanical, electrical, security, IT, AV, data, publishing services, commissioning materials management, operations and maintenance. These individuals are a resource for the project team to help interpret guidelines and requirements and are also included in Deliverable B and Deliverable C project reviews.

Patron Housing: Patron support facility located on some temple sites that provides all the same spaces as an Arrival Center as well as overnight patron accommodation. Some Patron Housing Buildings may include small apartments for temple leadership as well as a Distribution Center or Family History Center.

PBO: Presiding Bishopric Office, the second committee in the four-step approval process.

Phase 1: Describes the work required to achieve leadership approval for a project. Includes Review 1, Review 2 and Deliverable A.

Phase 2: Describes the design development, construction documentation and specification work required to prepare a project for construction.

Phase 3: Construction phase of a project through substantial completion, which includes Contract Administration responsibilities.

Project Accountant: Special Projects Department (Owner) finance project accountant responsible for the project. Member of Owner Project Team.

Project Manager: Special Projects Department (Owner) PM, project manager responsible for coordination of the entire project. Managing member of Owner Project Team.

Project Review: The first committee in the four-committee approval cycle. Project Review is made up of all Special Project Department Regional Directors as well as Senior Leadership to include the Managing Director and Design Director.

Quantity Survey: Assessing and quantifying materials. Labor and other resources needed for a construction project. The Quantity Surveyor who performs the survey provides a detailed report, often called a “Bill of Quantities,” outlining the estimated costs and quantities of various elements required for the project.

Record Set: The official archived set of documents for Owner use at the completion of the project. Record drawings are a combination of the as-built drawings along with all other documentation required to know, understand, and maintain the facility.

Review 1: Preliminary project proposal including a site plan and greyscale building massing.

Review 2: Final project proposal including finalized site plan, a photorealistic exterior rendering with details of entry and tower. Photorealistic interior renderings of all major spaces to include Entry, Brides, Grand Stair, Baptistry, Endowment Room, Celestial Room and Sealing Room. At the direction of the design team, multiple options may need to be provided for each rendering.

Site Design Manager: Special Project Department (Owner) SDM, manager of proper layout, design and specification of all site features and landscape elements. Member of Owner Project Team.

SPD: Special Projects Department, two divisions Design and Construction Services (DCS) and Temple Facilities Services Division (TFSD) collectively responsible for the design, construction and ongoing maintenance of physical facilities.

SPD Engineering Team: Structural, Mechanical, Electrical, Commissioning, AV, Security, IT and Data engineers, responsible for reviewing projects for adherence to Owner engineering requirements.

Specifications: Project manual consisting of written requirements for materials, equipment, construction systems, standards, and workmanship. They explain the scope of work, any materials that are to be used, the

methods of installation, and the quality of workmanship to be used, and are prepared during the design phase by design consultant, interior design consultant and engineers.

Start-up Kit: (SUK) Any set of items needed to start or initiate operation of a building, project or part of a project. These items are Owner provided and installed unless otherwise noted. Examples include the temple admin SUK, temple laundry SUK, etc.

Submittals: Items prepared by the contractor as indicated in the technical specifications for submittal to the design consultant, interior design consultant and engineers for review. The submittals are reviewed for conformance to the contract documents.

Substantial Completion: The stage of the project when the work or a designated portion thereof is sufficiently complete that it can be used for its intended purpose, The portion must be in accordance with the contract documents and expectations.

Temple Department: A separate department responsible for temple program development and implementation.

Temple Department Representative: A regional Owner reviewer assigned to a global region, responsible for program review and implementation for projects.

TFHEC: This is the third of four committees in the approval process. The Temple Family History Executive Committee, also known as Pre-TSC, consists of two to three members of the Quorum of the Twelve Apostles, a member of the Presiding Bishopric, the executive director of the Temple Department, a member of the General Relief Society Presidency, and leadership of both the Special Projects and Temple Departments.

TFM: Temple Facility Manager responsible for the day-to-day management of the completed temple and often serves as on-site Owner representation during construction.

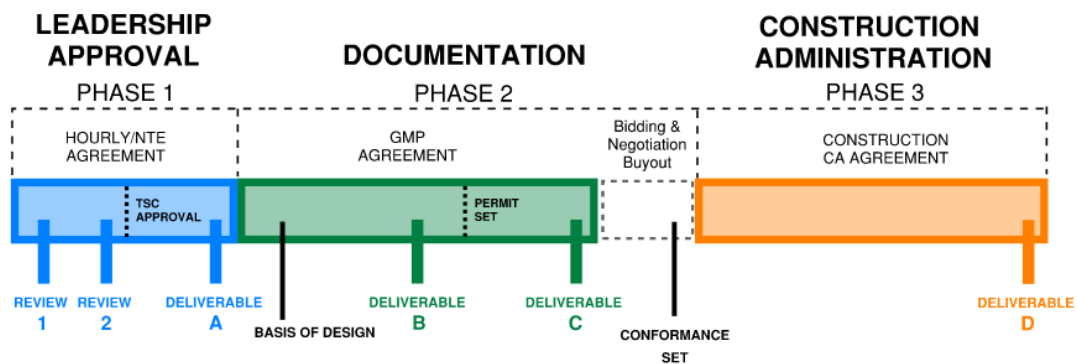
TFSD: Temple Facilities Services Division, the division of SPD responsible for the ongoing maintenance of physical facilities.

TMS: Temple Management System, also called PeopleSoft, is an Owner procurement system to issue POs to vendors, pay invoices and manage vendor orders.

TSC: Temple Sites Committee, synonymous with the “First Presidency” which represents the highest approval body in the church. This is the fourth and final committee in the approval process. TSC approved renderings

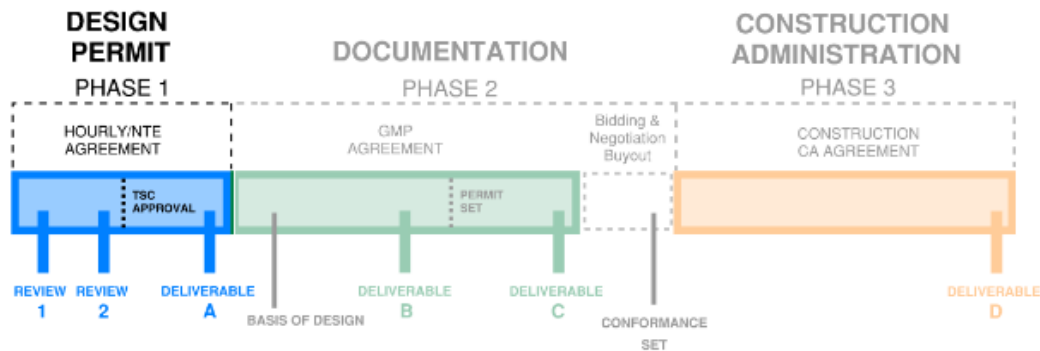
ii. PROJECT PHASES OVERVIEW:

Projects are divided into three phases representing the three main scopes of work of the project. Phase 1 “Leadership Approvals” includes work necessary to achieve an Owner approved design for Review 1, Review 2 and Deliverable A. This work is typically an hourly, not-to-exceed contract. Phase 2 “Documentation” includes the production of project documents to include drawings and specifications, with Owner reviews at the 50% and 100% completion milestones. This work is typically under a separate guaranteed maximum price contract. Phase 3 “Construction Administration” includes general oversight of the construction project, as well as providing specific direction and guidance to the General Contractor to ensure design intent is met. This work may be included in the guaranteed maximum price arrangement for Phase 2 or may be contracted separately in a specific CA agreement.



1A. PHASE 1 SUMMARY:

Provide architectural design compatible with the region, surrounding area and in keeping with the sacred nature of the building. Appropriately designed interior spaces which are correctly detailed for the intended use. The exterior and interior building aesthetics will not detract from or overshadow the patron's experience but complement and enhance the spiritual purpose of the building. Provide finishes and details of high-quality workmanship and materials, while minimizing long-term maintenance needs. Confirm materials meet quality standards for specified level of design and are in conformance with the Owner Temple Design Standards and Guidelines. Verify that design intent has been understood and coordinated between disciplines. Verify compliance with Owner Design Guidelines and Core Program. Provide contractual design services for Phase 1 to include Review 1, Review 2 and Deliverable A. Design consultants are responsible for managing hourly fees expended toward the not-to-exceed cap and notify the Owner Team, in advance, if additional funds are required. Work completed without prior approval may not be compensated.



1B. REQUIREMENTS:

1. **PRE-DESIGN KICK OFF MEETING:** Hold a Phase 1 pre-design kick-off meeting to establish team, schedule and milestones.
2. **BUDGET:** Provide design within the Owner's established project budget. Work and coordinate with Owner in preparing, reviewing, and maintaining design direction, design choices, and cost models that meet Owner budget.
3. **DESIGN ASSIST:** Coordinate with the Owner and Interior Design whether design assist contracts may be necessary in Phase 1 for stone selection and detailing, art glass design and constructability, decorative lighting, or other specialty items. If required, include that work and coordination in the overall schedule and as part of Deliverable A. Provide copies of design assist contracts to the Owner and include provisions for intellectual property ownership for design work developed.
4. **MATERIAL AVAILABILITY:** Ensure availability of fabrics, furniture, and interior finishes, in consultation with suppliers, before inclusion in project renderings. Every effort will be made during the development of the project, to confirm materials and finishes are locally or regionally available and constructable.

5. **MEETINGS:** Conduct and document regularly scheduled weekly project design meetings. Review schedule progress at each meeting, design updates, coordinate with any in-country resources for availability and suitability of regional materials. Keep Owner informed of decisions needed to progress according to schedule.
6. **MODELS:** Develop digital models, renderings, and presentation materials for items above for use in Owner approval meetings. Upload materials presented during meetings as a single, dated pdf or ppt into Owner project management system, as directed by Owner. Provide uploads immediately following weekly design meetings.
7. **SCHEDULE:** Provide a project schedule with weekly milestones to accomplish the deadlines in Phase 1.
8. **STANDARDS:** Ensure compliance with Owner furnished Standard Plan, Temple Program, Temple Design Guidelines and Standard Details, and other Owner Project Requirements (OPRs). Review standard plans for applicable program requirements, building codes and accessibility compliance.

1C. REVIEW 1 DELIVERABLE

SUMMARY:

Provide a regionally appropriate site plan and building massing options, informed by Owner provided due diligence and local precedent, for Owner review.

REQUIREMENTS: (alphabetical)

1. **ADDITIONAL VIEWS:** Provide additional site illustrations (Lumion or similar), as directed by SPD Owner Team, to demonstrate unique site features or views.
2. **CODE AND CULTURE REVIEW:** Provide preliminary code and zoning information to establish buildable envelope, setbacks, height restrictions, easements, accessibility and other local or regional requirements. Provide background research on the region, city, and site to include photos of regionally relevant architecture, historic sites, cultural sites, typical landscapes for public sites, color palette, climate data and any other information that will be helpful for design.
3. **COLORED SITE PLAN:** Provide a colored site presentation plan, utilizing Owner presentation standards and including all requested program space types. Show parking layout, driveways, outdoor utility and service areas, temple and ancillary buildings, walkways, plazas, monument sign(s), trees, shrub beds, lawn areas, and other significant site amenities.
4. **DESIGN STRATEGIES:** Every effort will be made during the development of site and massing options to confirm that design strategies are locally and regionally appropriate and constructable. Where possible utilize preapproved catalog massing options as a starting point, as directed by the Owner Design Team.
5. **MASSING OPTIONS:** Provide a minimum of two options for grey scale massings, shown from the same perspective view. Adjust as necessary to obtain Owner approval.
6. **PRECEDENT:** Research and precedent studies to inform architectural and interior design direction, including the following: images of regional exterior and interior architecture; regional flora and fauna, arts and culture to develop motifs, and to suggest appropriate furniture styles, fabrics, rugs, woods, stone, and other appropriate local materials. Provide preliminary concepts using local style precedent for architecture, interior architecture, details, finishes, furniture, fabric and accessories (use packages if applicable).
7. **PRESENTATION:** When site plan and massing options are acceptable to the SPD Owner Team, prepare high resolution images (jpegs), photoshop files (psd) and a presentation power point file (ppt) according to Owner standards and template, as described on the SPD Temple Design Hub SharePoint site under Presentation Standards, unless otherwise directed by the Owner. Revisions may be required at the SPD Owner Team's direction.

8. **SITE DESIGN:** Provide project site options that address location and orientation of required buildings, front or center spire options, hardscape and landscape features, site access and traffic flow, program or code required parking requirements (whichever is greater), and consideration of site services to include trash collection, mechanical enclosures, emergency access and bus routing.

SCHEDULE:

Review 1 deliverables will be completed within 1-2 months from project commencement, unless otherwise specified by the SPD Owner Team. This scope of work is considered complete when the project is approved by the Owner to move forward to Review 2.

1D. REVIEW 2 DELIVERABLE

SUMMARY:

Prepare exterior and interior photo-realistic rendered options to be presented to the Owner internal review process. Coordinate with Interior Design to provide proposals for art glass, rugs, deco lighting and deco paint for all rendered spaces for review.

REQUIREMENTS: (alphabetical)

1. **COLORED PLANS:** Provide colored floor plans for every level of the temple and ancillary buildings utilizing Owner presentation standards. Show finalized floor plans that have been reviewed for code and accessibility requirements. Show accurately sized furniture layouts that meet Owner program requirements.
2. **DESIGN:** A cohesive, finalized architectural and interior design with a coordinated color scheme, and design motifs. Provide an edited selection of final precedent images that have informed the design direction of the project to include regional and religious architecture, flora, color palette and motif inspiration.
3. **EXTERIOR RENDERINGS:** Develop photo-realistic renderings with regionally appropriate vegetation, trees, sky and backgrounds. Render to represent 12:00 noon on a sunny day, unless otherwise directed by SPD Owner Team. Provide views of temple front, rear, entry detail and tower/steeple detail, unless directed otherwise by Owner. Pay close attention to material representations to reflect approved and regionally available materials. Exterior renderings shall generally match the layout of Review 1 approved colored site plans. Submit high resolution jpegs as well as PSD, 3DMax and any revit files generated for renderings to the Owner.
4. **FINAL COLORED SITE PLAN:** Provide an updated site m-color plan utilizing Owner presentation standards. Show any required updates to the site plan, including number of parking stalls, number of steps, fences, plazas, site walls, raised beds, benches and other site amenities. Include sidewalks, buildings (including security, trash, utilities), specific fire lanes, surface retention basins, flagpole location and annual flower planting areas. Areas of special paving shall be graphically shown (paving pattern not required at this stage). Tree locations, shrub/groundcover, and lawn areas will be shown.
5. **INTERIOR DESIGN:** Coordinate with Interior Design for development of finishes to include decorative paint, art glass, rug and carpet development, decorative lighting, font and stair railing designs, decorative hardware and grilles, as well as any other items included in the interior design scope that directly impact Review 2 deliverables.

6. **INTERIOR RENDERINGS:** Provide photo-realistic renderings with regionally appropriate color palette, motif, furniture and accessory layout. Provide views of Temple Entry, Brides Room (if applicable), Grand Stair (if applicable), Baptistry, E Room, C Room and Sealing Room. Submit high resolution jpegs as well as PSD, 3DMax and any revit files generated for renderings to the Owner.
7. **MATERIAL AVAILABILITY:** Every effort will be made during the development of renderings, to confirm materials and finishes are locally or regionally available and constructable. Where possible utilize preapproved catalog finishes, and design templates as directed by the Owner Design Team.
8. **PLANS:** Coordinate closely with Interior Design to develop finalized, verified floor and furniture plans. Aesthetic space planning for furnishing and equipment required to comply with program, code and function. Provide reflected ceiling plans showing light fixture locations, sizes and types.
9. **PRESENTATION:** When exterior and interior renderings are acceptable to the SPD Owner Team, prepare high resolution images (jpegs), photoshop files (psd) and a presentation power point file (ppt) according to Owner standards and template, as described on the Temple CORE SharePoint site under Presentation Standards, unless otherwise directed by the Owner. Revisions may be required at the SPD Owner Team's direction.
10. **RENDERING PROCESS:** Present early draft renderings and make weekly updates based on Owner Team comments and program compliance. Review renderings with Owner Team a minimum of five times prior to finalization. Renderings are considered complete when Owner Team indicates they are ready for presentation. Renderings are considered final after First Presidency approval. Provide at least two design level options for Temple exterior and interior spaces, unless directed otherwise by SPD Owner Team.

SCHEDULE:

Review 2 deliverables will be completed within 2-3 months from Review 1 approval, unless otherwise specified by the SPD Owner Team. This scope of work is not considered complete until the project is granted final approval by the First Presidency. Once final approval has been granted, work on Deliverable A (schematic design) may commence.

1E. DELIVERABLE A:

SUMMARY:

The purpose of Deliverable A is to document decisions reflected by the rendered views. Items delineated below are to be provided in pdf format as well as a Revit model. Deliverable A drawing requirements will be architectural, (not m-color) and provided on a title block, drawn to scale, with gridlines and dimensions. Owner comments must be resolved before this phase will be considered complete.

REQUIREMENTS: (alphabetical)

1. **ARCHITECTURAL SITE PLAN:** Provide a site plan including site features shown in the approved colored plan.
2. **DELIVERY FORMAT:** Provide electronic files containing Revit Project model (LOD 200, see definitions), designs, drawings, and other items in both (1) PDF and (2) Revit project compatible format (RVT). Coordinate with Interior Design for submission of furniture, decorative lighting and other specific models.
3. **DOCUMENT REVIEW:** Upload PDF deliverable set into a review session for Owner review. Documents in sessions will be bookmarked and hyperlinked for full functionality. The session will be administered by the design consultant or by the SPD Design Manager, at the DM's discretion, with the Owner team having full access and permissions. The SPD Project Review Panel with Engineering Managers will be invited to provide comments over a two-week period, minimum. Design consultant will be required to respond to Owner comments on the review session and make corrections noted before this phase of work will be considered complete.
4. **DOOR HIERARCHY:** Provide door hierarchy for door and transom types, including finish and opaque/clear art glass locations.
5. **EXTERIOR ELEVATIONS:** Provide exterior elevations for all sides of all buildings, including materials and proposed heights.
6. **FLOORPLANS:** Provide floorplans for all buildings, all levels.
7. **FURNITURE PLANS:** Coordinate with Interior Design to provide furniture plans for all buildings, all levels.
8. **INTERIOR DESIGN:** Coordinate with interior design for development of deco paint, art glass, rug and carpet development, decorative lighting, railing design, decorative hardware and grilles, as well as general material selection for all spaces in the temple and ancillary building(s).

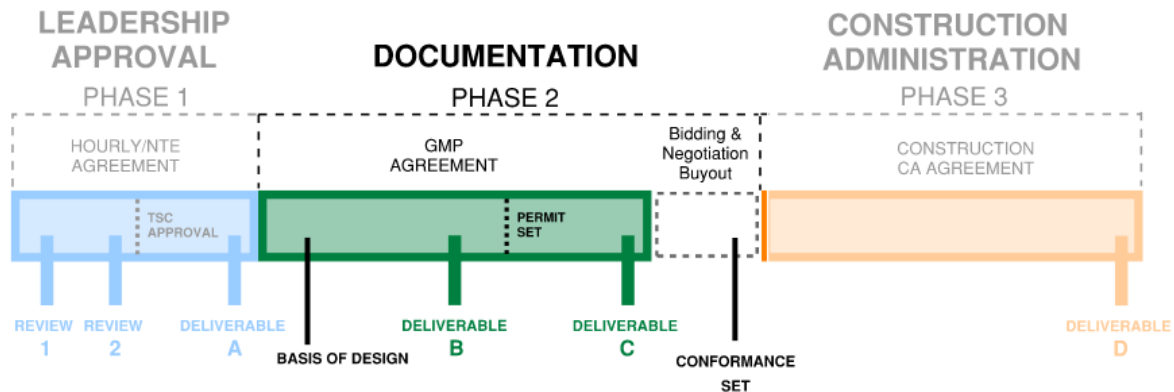
9. **INTERIOR ELEVATIONS:** Provide detailed interior elevations for all rendered rooms including all rendered design elements. Provide schematic interior elevations for all non-rendered rooms, in all buildings. Include doors, windows, ceilings, base and floors, at a minimum.
10. **PROGRAM AUDIT:** Utilizing the Owner live, shared Excel template, provide a comprehensive program audit of the building and interior spaces using the Owner provided program. Identify any areas of the plans that do not comply with program, exceed program or are not included due to space constraints. Specifically audit furniture for program compliance. Provide functional design considerations for better program integration where appropriate. Print a pdf copy of full program audit as well as a deficiency log. Include pdf in document upload.
11. **PROJECT INFORMATION PACKET:** Provide project information using an Owner provided template, describing design intent, motifs and material selections (preliminary Leadership Information packet).
12. **REFLECTED CEILING PLANS:** Coordinate with Interior Design to provide reflected ceiling plans for all buildings, all levels. Show decorative light fixture locations, can lights, ceiling tile, crown, plant-ons and decorative paint locations.
13. **REVIT MODEL:** Develop revit model(s) (LOD 200) including all components shown in approved renderings. The Revit model(s) shall also include the design development of all other spaces for all buildings that were not shown in the approved renderings. The model(s) shall include design elements necessary for proper documentation of the project, including, but not limited to, exterior cladding, ceiling designs, decorative painting, crown molding and base profiles, wainscoting, door and window trim, flooring patterns, art glass, stair railings, program required casework and cabinetry, furnishings, etc.
14. **SCHEMATIC GRADING PLAN:** Provide a schematic grading plan (and model if available) showing topography and resolved accessibility. Include finish floor elevations of all buildings and any site-specific critical spot elevations as well as general proposed contours.

SCHEDULE:

Deliverable A will be completed within 4-6 weeks following PBO approval, unless otherwise specified by the SPD Owner Team. Deliverable A scope is considered complete when all Owner comments from the review session have been resolved or agreed upon with the Owner to transition to the Phase 2 team for further study and resolution.

2A. PHASE 2 SUMMARY:

This phase of work expands on the approved design direction received at the end of Phase 1 to create a full set of construction documents and specifications, that match the approved design intent, incorporating regulatory code and zoning compliance, building systems coordination, improved and developed project design and thoughtful material selection. The phase of work begins with a Basis of Design agreement documenting system and material selections. This document guides the remainder of the project development. The project team will include structural, mechanical, electrical, plumbing and other necessary engineering subconsultants who develop building systems that meet Owner project requirements, with a focus on those that are both regionally appropriate and available, as well as able to be maintained in an efficient and timely manner. Work in these phases is reviewed by the Owner at least twice, first at 50% project completion (Deliverable B) and again at 100% completion (Deliverable C). Phase 2 is a multi-step process: First a Basis of Design document is established and approved by the Owner, informing further design development. Design progress is reviewed by the Owner Review Team when the project reaches the 50% completion milestone and documents are uploaded for Owner review. Deliverable C is considered complete when 100% documents are uploaded, with previous Owner comments resolved. Design consultant is responsible to manage fees expended toward and notify the Owner Team, in advance, if additional funds are required. Work completed without advanced approval may not be compensated.



2B. REQUIREMENTS:

1. **PRE-DESIGN KICK OFF MEETING:** Schedule and manage a Phase 2 Architectural and Engineering (AE) pre-design kick-off meeting with the presence of owner representatives to discuss project vision, initial design brief, Owner's design standards, localization/regulations plans, local sustainability, risk management, team planning, communication protocols, project goals, schedules and milestones.
2. **BUDGET:** Provide design within the established project budget. Work and coordinate with Owner in preparing, reviewing, and maintaining design direction, design choices, and cost models that meet Owner budget.

3. **DESIGN ASSIST:** Coordinate with the Owner and Interior Design whether design assist contracts may be necessary in Phase 1 for stone selection and detailing, art glass design and constructability, decorative lighting, or other specialty items. If required, include that work and coordination in the overall schedule and as part of the schematic design deliverables. Provide copies of design assist contracts to the Owner and include provisions for intellectual property ownership for design work developed.
4. **DESIGN MEETINGS:** Conduct and document weekly project design meetings with the Owner. Review schedule progress at each meeting, interior design updates, coordinate with any in-country resources for availability and suitability of regional materials. Keep Owner informed of decisions needed to progress according to schedule.
5. **ENGINEERING COORDINATION:** Conduct and document regular engineering coordination meetings to direct and guide the work of the project engineers to meet Owner Program and well as Guidelines. Regularly report to the Owner in design meetings, any major system decisions, coordination efforts and design development progress. Design consultant is responsible for ensuring architectural documents are coordinated with engineering.
6. **MATERIAL SELECTION:** Ensure availability of fabrics, furniture, and interior stones, in consultation with suppliers and stone design assist consultants, before inclusion in project renderings.
7. **OWNER REQUIREMENTS:** Ensure compliance with Owner furnished Standard Plan, Temple Program, Temple Design Guidelines and Core Details, and other Owner Project Requirements (OPRs). Review standard plans for applicable program requirements, building codes and accessibility compliance.
8. **OWNER REVIEWS:** Provide a progress set of construction documents at 50% documentation completion (Deliverable B) and 100% documentation completion. Set up and manage document review sessions to be reviewed by stipulated Owner representatives and other members of the project team. Provide written responses to each comment provided in reviews, revise construction documents as required and verify items are closed with the owner. Design progress cannot occur until owner review comments are addressed and closed.
9. **REVIT MODEL:** Develop revit models required for project coordination and BIM clash analysis. Upload materials presented during meetings as a single, dated pdf or ppt into Owner project management system, as directed by Owner. Provide uploads immediately following weekly design meetings.
10. **SCHEDULE:** Provide a week-by-week project milestone schedule to accomplish the Phase 2 deliverables including a Basis of Design representing proposed and approved building systems (architectural and engineering), Deliverable B (50% documentation completion), Deliverable C (100% project completion), and a conformed full set of construction documents and specifications.
11. **ENERGY MODELING:** Develop HVAC energy model identify three types of localized systems showing lowest energy efficient verses highest energy efficient, while including owner maintenance history data in calculations.

12. **LIGHTING MODELING:** Provide photometric and pseudo color renderings of lighting in the following areas, inside temple, temple exterior façade, site walkways and parking.
13. **ELECTRICAL UTILITY ANALYSIS:** Investigate and report to the owner the condition of the utility and conditioning and or protection proposals.
14. **RENEWABLE TECHNOLOGIES:** Propose commonly used renewable technologies in the region and provide a fee adjustment to Owner if renewable technology is selected.
15. **NEW TECHNOLOGY REQUEST:** Establish prior approval with owner concerning specified equipment/devices connecting to Owner network. All devices require prior Owner approval.
16. **ACOUSTIC REVIEW:** Design acoustic mitigation following Owner standards

2C. BASIS OF DESIGN (BOD) MILESTONE:

The Basis of Design document will follow the Owner template and reflect Owner requirements, proposed systems for consideration and major design considerations for the following systems: envelope, structure, windows, doors, interior finish materials, decorative lighting, millwork, emergency and backup systems, mechanical, electrical, plumbing, fire protection, renewable technologies and major code requirements. BOD Examples are provided on the Design Hub. This document will be submitted to the Owner team for review and comment. Final system selections will be documented and will guide the remainder of the project development.

2D. DELIVERABLE B:

SUMMARY:

Provide contract documents, drawings and specifications, representing 50% documentation completion. Deliverable B drawing requirements will be provided on a consistent title block, drawn to scale, with gridlines. The engineering work presented will be advanced enough for Owner review and comment. Information provided for this phase will comply with requirements for Temple Interior Design Manual Template. Pdf documents to be submitted and uploaded to the Owner Project Management platform. Establish a document review session and ensure the entire Owner team has full permission enabled. Documents in sessions will be bookmarked and hyperlinked for full functionality. Design consultant will be required to respond to Owner comments on the review session and make corrections noted before this phase of work will be considered complete.

REQUIREMENTS: (alphabetical)

1. **ACOUSTICAL:** Provide designs that minimize noise transfer between spaces and between site and neighboring properties per the Owner standards including noise reverberation. Includes modeling, material selection, sound insulation and masking, vibration control and reverberation control.
2. **AUDIO VISUAL:** This scope consists of devices, equipment, speakers, cabling, equipment room, projector, pathways, etc. Provide scaled/dimensioned plans for equipment rooms, rack placement, equipment layout and devices. Establish one-line drawings that define owner audio visual one-line standards, individual data/IP devices, room locations, data equipment and cable/device schedules. Incorporate Owner standards for facility management audio visual alerting and communication. See Owner standards for system selection type and performance.
3. **CIVIL:** Perform studies and provide reports identifying project feasibility, topographical survey, geotechnical survey, environmental impact, erosion and sediment control. Provide drawings that identify utility routings/connections, structural, mechanical/electrical infrastructure, site grading, storm water flow and management, hard and soft scaping, roadway and pavement.
4. **COMMUNICATION & DATA:** This scope consists of the main distribution frame (MDF), intermediate distribution frame (IDF), fiber, data/IP devices, equipment room, cabling, pathways, shafts and chases, etc. Provide scaled/dimensioned plans for equipment rooms, rack placement, equipment layout, devices, utility connection. Establish one-line drawings that define utility connection, MDF, IDF, fiber (strand, count & type), individual data/IP devices, room locations. Provide wireless access point heat mapping calculations. See Owner standards for system selection type and performance.
5. **DECORATIVE LIGHTING:** Coordinate with Interior Design to document decorative lighting fixture design and specifications. Show decorative lighting elevations that specify the correct scale and mounting height of fixtures. Provide 3D sample mockups and material samples and finishes for Owner approval. Create lighting schedule that includes type, quantity, location, ceiling height, drop, volts, lamp type/watts, actual lumens versus required lumens, weight (coordinate with structure and hoist

locations), manufacturer, model number and finish. Include decorative lighting performance in overall space lighting calculations. Include location of hoisting elements such as motor, electrical take-up reel, and safety arrestor.

6. **DEFERRED SUBMITTAL (TENDERING):** A schedule shall be established on the drawings, detailing the remaining design scope assigned to the Contractor. This schedule must ensure compliance with the Owners standards. Deferred submittals will be reviewed by the design consultant and the Owner SPD review team prior to approval.
7. **DEVICE COORDINATION:** Design consultant responsible to coordinate device locations with art locations, furniture, millwork details to include, but is not limited to: switches, outlets, fire extinguishers, fire alarms and annunciators, phone and data jacks, thermostats, decorative grilles, decorative lighting and sconce placement, sensors, control devices and access doors, diffusers, wifi devices, speakers, session clocks, decorative clocks, audiovisual equipment and sign placement. Coordinate devices on plans, reflected ceiling plans, and elevations regarding finishes, furnishings, lamps, art, mirrors, sconces, decorative paint locations and systems furniture. Review architectural/interior drawings with electrical and mechanical drawings to avoid discrepancies. This scope of work intentionally overlaps with architecture, coordinate with design consultant to ensure accuracy and quality control.
8. **DISTRIBUTION CENTER:** Provide space layout and design power, lighting, fire alarm, data/communication, intrusion, security, devices, equipment/lighting schedules, finishes and furnishings. Provide drawings that support the sales floor, aesthetic and security as defined in the Owner Standards. Use the provided Distribution Center component library:
9. **DOOR AND WINDOW SCHEDULES:** Provide window types, sizes, materials, manufacturer, glazing, hardware, fire rating requirements for window and door types utilized in the project. Provide detailed hardware specifications that comply with Owner Guidelines and keying requirements.
10. **DRAWINGS:** Provide plans, reflected ceiling plans, interior elevations, sections, schedules, tendering/deferred submittals schedules and details along with specifications, coordinated with engineering scopes to represent content items listed below. Work will be reviewed by design consultant to provide a high level of accuracy, coordination and quality control, prior to Owner submission. Project will thoroughly describe design intent and scope of work and provide information required for bidding, permitting and construction.
11. **ELECTRICAL:** This scope consists of power, distribution, lighting, fire alarm, devices, grounding, lightning protection, metering, pathways, automation, monitoring, equipment & lighting schedules, back up/storage, renewable energy, etc. Provide scaled and dimensioned plans with electrical equipment rooms, interior and exterior equipment, shafts, chases, pathways, utility connections, devices, lighting. Establish one-line drawings that define utility connection, room locations, equipment, circuitry protection, pathways, cabling, circuitry, circuitry loads, grounding, voltage drop calculations, calculated, demand estimates, connected, diversity factor loads, short circuit, utility contacts. Provide utility histories including 24-month quantity and length of outages, voltage and current swells and sags, as well as power factor allowance. Provide estimated and calculated demand, connected loads and diversity factor loads for back up, storage (and renewable technology if applicable) equipment. Provide local lighting restriction codes, local practices and project implementation plan on lighting sheets. See Owner

standards for system selection type and performance. Incorporate Owner standards for facility management intrusion alerting and communication. Provide lightning histories and codes on drawings. Provide fire alarm drawings that are consistent with local requirements and practices. Provide smoke detection if a fire protection design is not provided. See Owner standards for system selection type and performance.

12. **ENVELOPE:** Document design of exterior components of buildings to include walls, roofs, windows, door, foundations, waterproofing, thermal and moisture protection systems. Provide data analysis to support wall assembly details. Include required plans, sections and detailing to reflect scope, material selection, and industry best practices.
13. **EQUIPMENT:** Represent program required equipment in drawings and elevations. Provide specifications that comply with Owner guidelines and program requirements.
14. **FINISHES:** Develop and coordinate finishes with interior design to select materials and provide plans and specifications to address floor, ceiling and wall finishes. Coordinate finish schedules, legends, keynotes and specifications for correct material selection. Coordinate with Interiors to ensure relevant finish information is entered into Fohlio. Coordinate finish schedule, legends and furniture plans for correct placement and sizing.
15. **HARDWARE:** Use the Owner catalog for decorative hardware selection, where available. For other hardware needs such as lockers and partitions, make hardware selections using standard hardware from Owner approved sources and standards.
16. **INTRUSION & SECURITY:** This scope consists of devices, equipment, cabling, equipment room, cable, projector, pathways, off-site monitoring. etc. Establish one-line drawings that define individual data and IP devices, room locations, data equipment and cable and device schedules, cable classification and quantity, certificates and warranty requirements for cabling and cameras. Incorporate Owner standards for facility management intrusion alerting and communication. Provide active models and program system files that are used by design consultant, subconsultants and contractors. See Owner standards for system selection type and performance.
17. **MECHANICAL:** This scope consists of HVAC, temperature and moisture conditioning and reheat, plumbing, font, fire protection, piping, ductwork, system modeling. Provide scaled dimensioned plans for mechanical equipment rooms, interior and exterior equipment, shafts and chases. Provide duct locations and layout, sizes, cross sections and intersections, intake, exhaust louvers. Provide system automation, monitoring, equipment schedules with capacities and electrical requirements, etc. Provide HVAC heating and cooling design load calculations using modeling software that includes building load input data, individual room heating, cooling input data, energy efficiency calculations for part cooling loading conditions. Provide HVAC input data to include building square footage or square meters. Provide breakout of system components, rough order of magnitude costs, equipment initial cost, energy cost per square foot or square meter, maintenance cost per square foot or square meter, first replacement cost and portion of replacement, project escalation inflation rate, 40-year by year financial comparison breakout including graphs, local availability and equipment support. Provide energy report comparison of local HVAC systems and energy conservation measures, including assumptions and justification for selected measures over alternatives. Identify HVAC temperature control zones per Owner standards. Identify utility connections and provider requirements including water makeup composition, collection, water re-use benefits, jurisdiction requirements. Provide local energy

restriction codes, local practices and project implementation plan on mechanical sheets. See Owner standards for system selection type and performance.

18. **MILLWORK:** Use Owner recommended millwork packages where possible. Obtain millwork control wood samples for project and finish books, using Owner approved finish standards. Coordinate design and detailing running trim, millwork, doors, windows, stairs, railings, font, recommend desk, etc. Design millwork details for cabinetry including undercounter lighting where program required, sock distribution and return cabinetry in shoe storage areas including hooks in sock return drawer to hold mesh bags for laundry. Avoid complex custom trim, doors, guard rail and stair handle profiles. Simplify quantity of profile types where possible. Design and coordinate elevations to include base, crown, ceiling moldings, wall and ceiling paneling, chair rail, columns, pilasters, portals and corner guards.
19. **PROGRAM AUDIT:** Utilizing the Owner live, shared excel template, provide an updated comprehensive program audit of the building and interior spaces using the Owner provided program. Identify any areas of the plans that do not comply with the program, exceed program or are not included due to space constraints. Review non-compliant program items with the Owner for further direction and changes, as directed. Specifically audit furniture for program compliance. Provide functional design considerations for better program integration where appropriate. Print a pdf copy of full program audit as well as a deficiency log. Include pdf in document upload.
20. **REVIT MODEL:** Revit model (LOD 300) reflecting required components. In addition to other project requirements, provide revit model for furnishings, artwork, mirrors, sconces and decorative lighting. Submit these models to the Owner for inclusion in catalog resources for future projects.
21. **STRUCTURAL:** Ensure non-structural components are engineered following the Owner standards using the structural engineer. Provide drawings that include stairs accommodating building movement, edge of slab with cladding attachment details, floor and roof accommodating construction activities. Include the following in the structural calculations, wind, seismic, dead and live load being consistent with designs. Provide active models and program system files that are used by design consultants, subconsultants and contractors.
22. **SAMPLES:** Provide multiple samples or images of proposed materials, fabrics, furniture styles, ornamentations, etc. for Owner approval. Provide digital documents (using Template), along with physical samples of rendered room materials including finishes, fabrics, wood, stone and tile samples.
23. **SITE DEVELOPMENT:** Design to include plans, elevations, and other graphic information necessary to provide full design intent, including paving patterns and materials, site walls design, heights, patterns and materials, site furnishing designs, products, colors, and materials, planting design with plant types indicated. Site lighting and building flood lighting locations, types, and product information on landscape and site plans, monument sign design and location, nativity location shown.
24. **SPECIFICATIONS:** Specification will include relevant project sections in a draft format, in a side-by-side layout with English, the local language and reflect owner guidelines and standards. Specifications will be tailored or tropicalized to the specific region where the project is located. Country or regional codes and standards will be met. Where local codes and standards are deficient or not available, comparisons with similar US Standards can be made by recommendation from the design consultant team for Owner approval. Major deviations or differences between the two will be reviewed with the

Owner for further direction. Provide specifications for specified finishes, systems, equipment, sequence of operations, studies, tests, surveys, reports, trainings, attic stock, spare parts, warranties, record documents. Review specifications related to finishes including paint, for accuracy in sheen and color. Review Architectural Woodwork Institute (AWI) or equivalent local requirements for premium-grade wood specifications, stone specifications, etc. Obtain MSDS sheets on interior products to include product specifications, maintenance and warranties.

25. **STONE SELECTION:** Use only Group A stones, generally defined as: Sound with uniform and favorable working qualities containing no geological flaws or voids. They include completely metamorphosed (stone), in which impurities such as clays and silts have reacted chemically with calcite or dolomite to form other minerals. These stones have uniform working qualities, can be used on the exterior or interior and do not require any filling or patching. Conform with Owner requirements for slip resistance and abrasion testing as indicated in the Design Standards and Guidelines. Demonstrate chain of custody for stone used; do not use stone where chain of custody cannot be demonstrated. Cut-to-size suppliers will mark slabs with block information and mark blocks with quarry and quarry layer information. Following this procedure will allow for procurement of the selected stone package in time for installation, conforming with project schedule. Provide stone specifications and comply with Owner performance measures.
26. **STONE & TILE:** Coordinate with Interior Design to select restroom tiles for walls and flooring using running line product. Select countertops and stone base for areas adjacent to cabinetry in wet areas. Provide details for door and jamb sections, plinth and threshold details, finish transition of threshold, casings, doors, cased openings, crown and trim moldings. Show stone threshold locations in areas where flooring material changes occur. Cross-reference details and ensure transitions are accounted for. Coordinate window mullion and muntin profiles, windowsill and aprons. At a minimum, provide stone or solid surface sills in wet areas. Provide stone and tile details for font surround, interior of font, tile to stone transitions, wet paths, stair finishes, stone cap and skirt around font opening and transitions at railing. Provide proposed tile patterns for owner approval.

SCHEDULE:

Deliverable B will be complete 4-5 months following Phase 2 commencement, unless otherwise specified by the SPD Owner Team. Deliverable B is considered complete when 50% contract documents are uploaded to the Owner for review and all Owner comments have been resolved or agreed upon with the Owner to transition to the Deliverable C phase for further study and resolution.

2E. DELIVERABLE C:

SUMMARY:

Provide contract documents, drawings and specifications, representing 100% documentation completion. Deliverable C drawing requirements will be provided on a consistent title block, drawn to scale, with gridlines. Information provided for this phase will comply with requirements for Temple Interior Design Manual Template. Pdf documents to be submitted and uploaded to the Owner Project Management platform. Establish a document review session and ensure the entire Owner team has full permission enabled. Documents in sessions will be bookmarked and hyperlinked for full functionality. Design consultant will be required to respond to Owner comments on the review session and make corrections noted before this phase of work will be considered complete.

REQUIREMENTS: (alphabetical)

1. **ACOUSTICAL:** This scope consists of modeling, material selection, sound insulation and masking, vibration control, reverberation control. Provide designs that minimize noise transfer between spaces and property site per the Owner standards including noise reverberation.
2. **AUDIO VISUAL:** This scope consists of devices, equipment, speakers, cabling, equipment room, projector, pathways, etc. Provide scaled/dimensioned plans for equipment rooms, rack placement, equipment layout and devices. Establish one-line drawings that define owner audio visual one-line standards, individual data/IP devices, room locations, data equipment and cable/device schedules, cable classification, quantity, certificates and warranty requirements. Incorporate Owner standards for facility management audio visual alerting and communication.
3. **CIRCULATION:** Provide plans and sections to illustrate vertical circulation, stairs and elevators, without conflict with windows, doors and ceilings.
4. **CIVIL:** Perform the following studies and reports identifying, project feasibility, topographical survey, geotechnical survey, environmental impact, erosion and sediment control. Provide drawings that identify utility routings/connections, structural, mechanical and electrical infrastructure, fire protection and access, site grading, storm water flow and management, hard and soft scaping, vegetation coordination, roadway and pavement.
5. **CLADDING:** Provide complete documentation for proposed cladding systems to include integration with envelope materials, structural engineering supports and coordination, elevations with dimensions and details. Detail stone cladding using industry best practices and in coordination with a stone design assist consultant, where available. Detail precast, GFRG and other exterior cladding systems according to industry best practices and ASTM and Natural Stone Institute standards.
6. **COMMUNICATION & DATA:** This scope consists of the main distribution frame (MDF), intermediate distribution frame (IDF), fiber, data/IP devices, equipment room, cabling, pathways, shafts and chases, etc. Provide scaled/dimensioned plans for equipment rooms, rack placement, equipment layout, devices,

utility connection. Establish one-line drawings that define utility connection, MDF, IDF, fiber (strand, count & type), individual data/IP devices, room locations, data equipment and cable/device schedules, cable classification, quantity, certificates and warranty requirements. Provide wireless access point heat mapping calculations. Provide active models and program system files that are used by design consultant, subconsultants and contractors.

7. **DECORATIVE LIGHTING:** Coordinate with Interior Design to document decorative lighting fixture design and specifications. Show decorative lighting elevations that specify the correct scale and mounting height of fixtures. Provide 3D sample mockups and material samples and finishes for Owner approval. Create lighting schedule that includes type, quantity, location, ceiling height, drop, volts, lamp type/watts, actual lumens versus required lumens, weight (coordinate with structure and hoist locations), manufacturer, model number and finish.
8. **DEFERRED SUBMITTALS (TENDERING):** A schedule shall be established on the drawings, detailing the remaining design scope assigned to the Contractor. This schedule must ensure compliance with the Owners standards.
9. **DEVICE COORDINATION:** Coordinate device locations with art locations, furniture, millwork details to include, but are not limited to: switches, outlets, fire extinguishers, fire alarms and annunciators, phone and data jacks, thermostats, decorative grilles, decorative lighting and scone placement, sensors, control devices and access doors, diffusers, wifi devices, speakers, session clocks, decorative clocks, audiovisual equipment and sign placement. Coordinate devices on plans, reflected ceiling plans, and elevations regarding finishes, furnishings, lamps, art, mirrors, sconces, decorative paint locations and systems furniture. Review architectural/interior drawings with electrical and mechanical drawings to avoid discrepancies. This scope of work intentionally overlaps with architecture, coordinate with design consultant to ensure accuracy and quality control. Illustrate typical mounting heights for switches, receptacles, alarm devices, card readers, thermostats, session clocks and other wall-mounted devices. Ensure alignment and proper spacing that does not intrude on high-priority ordinance aesthetics.
10. **DISTRIBUTION CENTER:** This scope consists of power, lighting, fire alarm, data, communication, intrusion, security, devices, equipment, lighting schedules, finishes and furnishings. Provide drawings that support the sales floor, aesthetic and security as defined in the Owners Standards.
11. **DOOR AND WINDOW SCHEDULES:** Provide window types, sizes, materials, manufacturer, glazing, hardware, fire rating requirements for window and door types utilized in the project. Provide detailed hardware specifications that comply with Owner Guidelines and keying requirements.
12. **DRAWINGS:** Provide plans, reflected ceiling plans, interior elevations, sections, schedules and details along with specifications, coordinated with engineering scopes. Work will be reviewed by design consultant to provide a high level of accuracy, coordination and quality control, prior to Owner submission. Project will thoroughly describe design intent and scope of work and provide information required for bidding, permitting and construction.
13. **ELECTRICAL:** This scope consists of power, distribution, lighting, fire alarm, devices, grounding, lightning protection, metering, pathways, automation/monitoring, equipment and lighting schedules, back-up storage, renewable energy, etc. Provide scaled, dimensioned plans for electrical equipment rooms, interior, exterior equipment, shafts, chases, pathways, utility connections, devices, lighting. Establish one-line drawings that define utility connection, room locations, equipment/circuitry protection, pathways,

cabling, circuitry, circuitry loads, grounding, voltage drop calculations, calculated, demand estimates, connected, diversity factor loads, short circuit and utility contacts. Provide utility histories with 24-month quantity and length of outages, voltage and current swells and sags and power factor allowance. Provide dedicated estimated calculated, demand, connected, diversity factor loads for back up, storage equipment and renewable technology if applicable. Provide arc flash and coordination studies for the distribution system. Provide shop drawing level lighting control, if applicable per Owner standards. See Owner standards for system selection type and performance. Provide photometric studies and pseudo color renderings for interior and exterior lighting that include light level measurements. Provide local lighting restriction codes, local practices and project implementation plan on lighting sheets. Provide shop drawing level system design for Building Management System (BMS) or Equipment Monitoring System (EMS). See Owner standards for system selection type and performance. Incorporate Owner standards for facility management intrusion alerting and communication. Provide active models and program system files that are used by design consultant, subconsultants and contractors. Provide lightning histories and codes on drawings. Provide fire alarm drawings that are consistent with local requirements and practices. Provide smoke detection if a fire protection design is not provided.

14. **ENGINEERING COORDINATION:** Coordinate the work of project engineering scopes with architecture and ensure and verify elimination of conflicts and discrepancies. Include civil engineering, landscape, structural, mechanical, electrical, plumbing, fire, envelope, as well as any others as directed by the Owner.
15. **ENVELOPE:** Design consultant responsible for documenting the design of exterior components of buildings to include walls, roofs, windows, doors, foundations, water-proofing and damp proofing systems. Include required plans, sections and detailing to reflect scope, material selection, and industry best practices. Coordinate with an envelope consultant as directed by the Owner team.
16. **EQUIPMENT:** Represent program required equipment in drawings and elevations. Provide specifications that comply with Owner standards and guidelines and program requirements. Provide early bid packages for long lead items like elevators or mechanical equipment, as directed by the Owner.
17. **FINISHES:** Finalize coordination of finishes with interior architecture to select materials and provide plans and specifications to address furniture, art plans, decorative mirror locations and designs, accessory types and locations and fabric specifications including testing requirements. Coordinate finishes and order a minimum of three samples from manufacturers. Coordinate metal finishes in the temple to include door hardware, lighting fixtures, grilles and font railing. Coordinate finish schedule, legends and furniture plans for correct placement and sizing.
18. **FOHLIO:** Coordinate with Interior Design to ensure relevant project information is updated and complete in Fohlio.
19. **HARDWARE:** Use the Owner catalog for decorative hardware selection, where available. For other hardware needs such as lockers and partitions, make hardware selections using standard hardware from Owner approved sources and standards.
20. **INTRUSION & SECURITY:** This scope consists of devices, equipment, cabling, equipment room, cable, projector, pathways, offsite monitoring. etc. Establish one-line drawings that define individual data, IP devices, room locations, data equipment and cable and device schedules, cable classification and quantity, certificates and warranty requirements for cabling and cameras. Identify local offsite 24-hour monitoring agency and approve use with owner. Incorporate Owner standards for facility management intrusion

alerting and communication. Provide active models and program system files that are used by design consultant, subconsultant and contractors.

21. **LANDSCAPE:** Further define all information provided in Deliverable B. Provide and coordinate with other disciplines all drawings, specifications, and other necessary items for bidding, permitting, and construction, and meet Owner program according to PM direction. Include irrigation plans where requested. Provide a sample board and sample of proposed products and finishes to be used. Design consultant shall also start preparation and submittal of draft Temple Site Management Plan.
22. **MECHANICAL:** This scope consists of HVAC, temperature and moisture conditioning and reheat, plumbing, fire protection, piping, ductwork, system modeling. Provide scaled dimensioned plans for mechanical equipment rooms, interior and exterior equipment, shafts and chases. Provide duct location and layout, sizes, cross sections and intersections, intake/exhaust louvers. Provide system automation, monitoring, equipment schedules with capacities and electrical requirements, etc. Provide HVAC heating and cooling design load calculations using modeling software that includes building load input data, individual room heating/cooling input data, energy efficiency calculations for part cooling loading conditions. Provide HVAC input data to include building square footage or square meter. Provide breakout of system components, rough order of magnitude costs, equipment initial cost, energy cost per square foot or square meter, maintenance cost per square foot or square meter, first replacement cost and portion of replacement, project escalation inflation rate, 40-year by year financial breakout including graphs, local availability and equipment support. Energy report comparison of local HVAC systems and energy conservation measures, including assumptions and justification for selected measures over alternatives. Identify HVAC temperature control zones per Owner standards. Identify utility connections and provider requirements including water makeup composition, collection, water re-use benefits, jurisdiction requirements. Provide font pump flow rates, font water level and distance related to pump and identify projected net positive suction head available (NPSHA)/Net positive suction required (NPSHR). Provide air and water flow diagrams showing in cubic feet per minute and gallons per minute. Provide local energy restriction codes/ local practices and project implementation plan on mechanical sheets. Provide shop drawing level system design for Building Management System (BMS) or Equipment Monitoring System (EMS). See Owner standards for system selection type and performance. Provide active models and program system files that are used by design consultant, subconsultants and contractors.
23. **MILLWORK:** Provide project required millwork with complete detailing that complies with guidelines and program. Specify materials, both solids and veneers, finish, grain, and layup patterns. Use Owner recommended millwork packages where possible. Provide final project reference samples for species and finishes specified on the project, using Owner approved finish standards.
24. **PROJECT INFORMATION PACKET:** Update the previous project information for the Groundbreaking ceremony using an Owner provided template, describing design intent, motifs and material selections (preliminary Leadership Information packet).
25. **PROGRAM AUDIT:** Utilizing the Owner live, shared excel template, provide an updated comprehensive program audit of the building and interior spaces using the Owner provided program. Identify any areas of the plans that do not comply with program, exceed program or are not included due to space constraints. Specifically audit furniture for program compliance. Provide functional design considerations for better program integration where appropriate. Print a pdf copy of full program audit as well as a deficiency log. Include pdf in document upload. Program items will be resolved and complete or noted as non-compliant and agreed to by the Owner.

26. **REVIT MODEL:** In addition to other project requirements, provide revit model for furnishings, artwork, mirrors, sconces and decorative lighting. Submit these models to the Owner for inclusion in catalog resources for future projects.
27. **SAMPLES:** Provide any necessary updates to samples or images of proposed materials, fabrics, furniture styles, ornamentations, etc. for Owner approval. Provide digital documents using Owner template, along with physical samples of rendered room materials including finishes, fabrics, wood, stone and tile samples.
28. **SIGNAGE:** Provide building signage elevations and details to include exterior stone inscriptions (monument sign, cornerstone and building inscription) as well as interior room signage. Signage proposal packages require separate Owner approval prior to construction.
29. **SPECIFICATIONS:** Specification will include relevant project sections in a draft format, in a side-by-side layout with English and the local language. Specifications will be tropicalized to the specific region where the project is located. Country or regional codes and standards will be referenced and compared with similar US Standards. Major deviations or differences between the two will be reviewed with the Owner for further direction. Provide specifications for specified finishes. Provide specifications for specified finishes, systems, equipment, sequence of operations, studies, tests, surveys, reports, trainings, attic stock, spare parts, warranties, record documents. Consolidate warranties, training, tendered/deferred submittals, attic stock, spare parts into separate specification document for owner review. Review specifications related to finishes including paint (for accuracy in sheen and color). Review Architectural Woodwork Institute (AWI) or equivalent local requirements for premium-grade wood specifications, stone specifications, etc. Equipment/system performance, expected maintenance, life expectancy, system sequence of operations, commissioning (include Owner's commissioning plan) Obtain MSDS sheets on interior products to include product specifications, maintenance and warranties.
30. **STONE & TILE:** Select restroom tiles for walls and flooring using running line products. Select countertops and stone base for areas adjacent to cabinetry in wet areas. Provide details for door and jamb sections, plinth and threshold details, finish transition of threshold, casings, doors, cased openings, crown and trim moldings. Show stone threshold locations in areas where flooring material changes occur. Cross-reference details and ensure transitions are accounted for. Coordinate window mullion and muntin profiles, windowsill and aprons. At a minimum, provide stone or solid surface sills in wet areas. Provide stone and tile details for font surround, interior of font, tile to stone transitions, wet paths, stair finishes, stone cap and skirt around font opening and transitions at railing. Provide proposed tile patterns for owner approval.
31. **STRUCTURAL:** Ensure non-structural components are engineered following Owner standards using the structural engineer. Provide drawings that include stairs accommodating building movement, edge of slab with cladding attachment details, floor and roof accommodating construction activities. Include the following in the structural calculations, wind, seismic, dead and live load being consistent with designs. Provide active models and program system files that are used by design consultant, subconsultants and contractors.
32. **WALL SYSTEMS:** Provide sections, schedules and details accurately describing wall systems, types, materials, assemblies, acoustical ratings that meet Owner requirements. Ensure acoustic separation in ordinance spaces per guidelines.

SCHEDULE:

Deliverable C will be complete 4-6 months following Deliverable B, unless otherwise specified by the SPD Owner Team. Deliverable C is considered complete when 100% contract documents are uploaded to the Owner for review and final Owner comments have been incorporated into drawings and specifications.

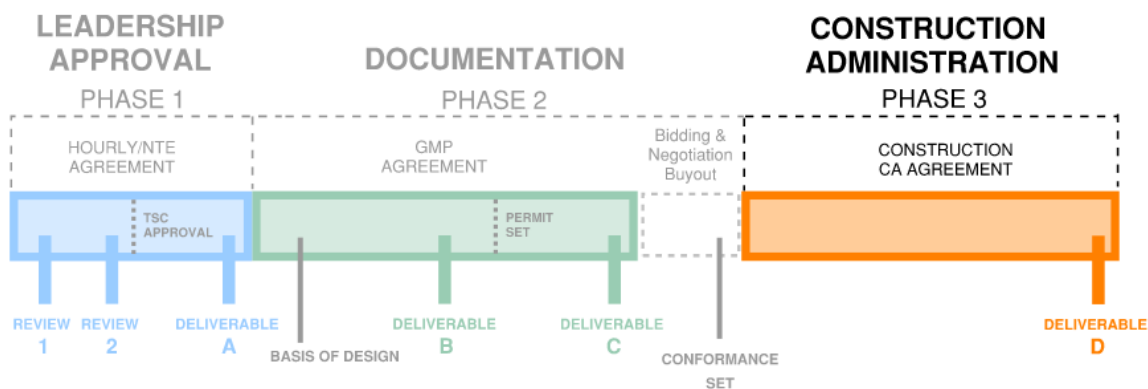
2F. CONFORMED SET MILESTONE:

The Owner may require a final Conformed Set of documents (drawings and specifications) following the bidding and negotiation process. A conformed set will incorporate updates, changes and additions collected from Requests for Information (RFIs) and any Addenda issued during bidding. The Conformed set of drawings and specifications is typically referred to as the official Construction Documents for the Construction Administration process. The Owner team will set a deadline for the submission of Conformed Set based on the completion of the Bidding and Negotiation period. Coordinate with Interior Design to include any interior specific information required.

3A. PHASE 3 SUMMARY:

Provide Construction Administration services and project oversight to facilitate the construction process and ensure the design intent of the documentation is followed by the General Contractor. Oversee construction as indicated below. Provide regular communication and updates for the Owner about construction status, issues requiring Owner direction and proposed solutions. Coordinate with Interior Design to provide post-installation support, communicating with the team about required follow-up and any changes that take place during install or after Temple Department inspection. Coordinate with Interior Design to provide completed O&M manuals and other project closeout documents for all buildings associated with the site, including as-built documentation.

Phase 3 includes construction contract administration services and oversight for the implementation of design intent during the duration of the construction of the project. The project is not considered complete until the final closeout project deliverables are received by the Owner (Deliverable D). Design consultant is responsible to manage fees expended toward and notify the Owner Team, in advance, if additional funds are required. Work completed without advanced approval may not be compensated.



3B. REQUIREMENTS: (alphabetical)

1. **ADMINISTRATION:** Monitor the construction project, for the purposes of ensuring the construction is executed according to the design intent described by project drawings and specifications. Attend regular Owner-Architect-Contractor (OAC) meetings. As part of Administration duties monitor construction progress to validate pay applications, including the following verifications and quality control measures:

Compliance: Verify construction complies with approved plans, codes and regulations

Work Quality: Assess completed work quality and ensure it meets specifications and design intent

Progress: Monitor progress of construction against the project schedule

Materials: Check quality and appropriateness of materials used; stored material conditions

Issue Resolution: Identify project issue assignments/responsibilities and track completion

2. **COMMISSIONING:** Report and track the commissioning functional testing and general material/installation testing issues, including the correction of issues to Owner. Provide correction recommendation through the issue management processes. Include the Owner commissioning

representative in OAC meetings. Provide final commissioning issues completion verification to owner. Oversee and manage construction change orders, ensuring accurate documentation, assessment of impacts on cost and schedule, and timely communication of progress and resolutions to the owner.

3. **FIELD REPORTS:** Provide and manage report submissions as follows:

Project Information: Basic details such as project name, location, and reference ID
Observation Date and Time: Specific date and time of the site visit
Weather Conditions: Current weather conditions, adverse/severe weather impacting schedule
Attendees: List of individuals present during the site visit
Construction Progress: Status of construction activities, milestones achieved, and work remaining
Quality of Work: Assessment of workmanship and adherence to design specifications
Material and Equipment Status: Condition and availability of materials and equipment on site
Compliance Check: Verification of compliance with plans and Owner standards
Issues and Resolutions: Identification of issues and steps taken or recommended to resolve
Photographic Evidence: Categorized photos documenting the current state of the project
Recommendations: Suggestions for improvements or necessary actions
Next Steps: Outline of upcoming tasks and activities
Correction Review: Track correction items until complete
Information Access: Consolidated reports with bookmarked PDFs for ease of future reference

4. **PUNCHLIST:** Review the Contractor prepared punch list for accuracy and completeness. Ensure completed work matches the design intent of contract documents.
5. **RFI/SUBMITTALS/SHOP DRAWINGS:** Review and track construction submittals on project finishes, systems, equipment, etc for all specifications sections requiring submittals. Track program conformity and design intent alignment through submittal reviews, correcting items that do not accurately reflect the intent expressed in contract documents or Owner standard requirements. Ensure the remaining design scope intended to be included in deferred submittals/tendering are detailed and assigned to the Contractor on the drawings. Verify tendering/deferred submittal schedule responsibilities are completed and fully reviewed for conformance with design intent.
6. **SITE OBSERVATIONS:** Visit the site regularly, (number and frequency of visits specified per contract) to ensure interior Architectural finishes and materials are installed as specified. Ensure proper installation of devices on walls and ceilings prior to sheetrock installation. Monitor construction progress for program conformity. Visit suppliers if requested by the Owner. Visit storage site or warehouse as requested to verify materials meet specifications and contract documents. Document visits with a field report, provided to Owner, identifying any items that require correction. Use the criteria listed for field reporting. Require sub-consultants to visit and provide site observations as specified by contract.
7. **TRAVEL:** Visit the site as requested by the Owner and specified by contract, to ensure Architectural finishes and materials are installed as specified and the design intent is being met. Visit suppliers as requested by the Owner.
8. **RISK MANAGEMENT:** Identify potential risks and conflicts in the design and develop strategies to mitigate these risks. Assess the impact of identified risks and coordinate mitigation measures with all stakeholders. Prepare contingency plans, with all stakeholders, for identified risks to ensure the project can continue in the event of unforeseen issues.

3C. DELIVERABLE D:

SUMMARY:

Provide closeout documents and project manuals as outline below. Deliverable D requirements will comply with Owner Guidelines and Standards. All pdf documents are to be submitted and uploaded to the Owner's Project Management platform. Physical books will be delivered to the location indicated by the Owner team. Final delivery of all required items will be required for this final phase of work to be considered complete.

REQUIREMENTS: (alphabetical)

1. **AS-BUILT DOCUMENTS:** During Construction Administration, changes to the documents will be clouded and referenced in official project documentation by the General Contractor. The GC will provide these documents within 60 days of substantial completion to the design consultant. Provide an updated revit model (LOD 400) reflecting as-built conditions.
2. **PROJECT INFORMATION PACKET:** Update the previous project information for the Dedication using an Owner provided template, describing design intent, motifs and material selections (preliminary Leadership Information packet).
3. **PROJECT MANUALS AND CLOSEOUT:** Utilize the Interior Design Manual template in order to create project manuals Provide project pages for furniture, fabrics, finishes, manufacture, finish color. Accessory (better description). Temple and any Ancillary buildings. Provide the Temple Site Management Plan, updated Leadership Information packet and a Dedication Report.
4. **RECORD DOCUMENTATION:** The design consultant is to provide certified Record Drawings generated from the As-Built drawings submitted by the General Contractor, in combination with project addenda and design revisions. Record drawings will be stamped and marked by the design team as "Record Drawings" and dated prior to submission. In addition, provide collated and organized Operations and Maintenance manuals and required training in both paper and electronic means as outlined in each individual product specification. Provide warranty summaries and installation manuals, testing reports, technology username and passwords, project photographs/images, and certificate of occupancy. An updated final revit model (LOD 400) will be provided, matching Record Drawings. Provide design model files, calculation/energy model files, and system program files.

SCHEDULE:

Deliverable D will be completed within 60 days of completion of furniture installation, unless otherwise specified by the SPD Owner Team.

4A. CODE & CULTURE CONSULTANT: PHASE 1

SUMMARY:

The code and culture consultant serves as a liaison between the primary design consultant firm and the Owner, to provide expert guidance on culture, climate regionally available materials as well as zoning, codes, ordinances and permitting process. As directed by the Owner, consultant facilitates any additional due diligence work on the property that may be required.

REQUIREMENTS: (alphabetical)

1. **ACCESSIBILITY REVIEW:** Review the Design Team's parking layout and site plan to ensure harmony with the overall site concept and local standards. Provide expert guidance on accessibility, including the location of passenger drop-off points and potential additional reviews in jurisdictions with restrictive accessibility requirements.
2. **CLIMATE REVIEW:** Provide input on local climatic and micro-climate conditions, weather patterns, humidity, and precipitation, to inform design decisions and material selection. Confirm if other environmental assessments are appropriate.
3. **CODES & ORDINANCE COMPLIANCE:** Confirm the relevant municipality or territorial authorities having jurisdiction over the site, including any indigenous people planning documents requiring consideration. Review and ensure compliance with all applicable code requirements, building codes, ADA regulations, zoning regulations, fire safety and other local ordinances or restrictions. Provide guidance on code issues such as setbacks, easements, height restrictions, planning and zoning requirements, permit process, and consult with authorities, as directed by Owner to determine project feasibility. Provide guidance to the Owner on whether additional services will be considered for code expeditors or permit facilitators based on the timeline and local permitting process.
4. **CULTURAL AND REGIONAL CONSIDERATIONS:** Consider the cultural identity, traditions, history, heritage, cultural sites and significant art, architecture, and landscape architectural features of the city, country, and region where the project is located. Provide input on local cultural practices and any indigenous religious or historical items of significance to avoid, negative local connotations or difficult historical narratives (e.g., avoiding certain motifs or patterns with sensitive historical associations).
5. **LOCAL MATERIALS:** Identify and help vet appropriate local and regional suppliers, fabricators, and materials. Assist the team in ensuring the regional availability of exterior and interior finishes, including stone, wood, fabrics, and furniture. As directed by Owner, work with an Owner-provided photographer to obtain drone and detailed photos of the site and surrounding area to support design decisions.
6. **MEETINGS:** Participate in and document regularly scheduled project design meetings, providing input on cultural and code elements, and ensuring that all materials presented are uploaded to the Teams library immediately following meetings.
7. **SITE REVIEW:** Review the site survey and property to confirm the presence of protected trees or vegetation and whether a tree survey or other assessments are appropriate. Review floodplains,

seismic zones, archaeological impacts and any other existing parameters that will be considered. Make recommendations for traffic study, additional geotechnical explorations, and existing utility locations. Confirm any history of site contamination or hazard to remediate as well as any adjoining sensitive uses that will be considered.

SCHEDULE:

Code and Culture consultancy work is considered complete when the primary design consultant has submitted Deliverable A to the Owner for review and approval. Project commencement to Phase 1 completion is typically a 4 – 6 month timeframe.

4B. CODE & CULTURE CONSULTANT: PHASE 2 - PERMITTING

SUMMARY:

In Phase 2, the code and culture consultant serves as the liaison between the primary design consultant firm and the authority having jurisdiction, to obtain all necessary project permits.

REQUIREMENTS:

1. **CORRESPONDANCE:** As required throughout the planning preparation, submission and approval process, ongoing correspondence and follow-up communication with the local municipality until all planning approvals are obtained.
2. **DOCUMENTS:** Determine all required documentation for Planning Permissions/Permitting and under the direction of the SPD Project Manager, gather existing documentation and arrange for any additional items required. Consider the following possible requirements: architectural and landscape plans, survey, traffic assessment, civil plans to include stormwater management, acoustic and visual impact assessments, engineering report addressing earthwork, site services and geotechnical requirements, soil contamination assessment, environmental assessment especially for removal of plants or trees, cultural and heritage impact assessment, sustainability assessment, demolition and waste management plans and detailed cost estimate.
3. **MEETINGS:** Meet with the Owner team and project design team as well as individual consultants as needed to facilitate preparation of the documentation required for permitting approval. Attend meetings with local municipality and key project stakeholders and public meetings that may be required as part of the notification process.
4. **PLANNING SUBMISSIONS:** Prepare and or manage the preparation from other consultants, all necessary documentation, reports and information necessary for all required formal planning submissions to the local municipality.
5. **PUBLIC NOTIFICATION:** If a public notification process is required, prepare all necessary planning submissions, required meetings with consultants, local municipality and others, prepare required evidence for hearings, attend all meetings, hearings and proceedings as required to obtain final planning approvals.
6. **SCHEDULE:** Establish, regularly update and meet a schedule for the overall planning preparation, submission and approval process. Identify key milestone dates for consultant engagement, completion of reports and planning documentation, formal planning submission and estimated planning approvals.

SCHEDULE:

This scope of work is considered complete when all required permits and approvals for a project have been obtained from the Authorities having jurisdiction. Permit timeframes vary from region to region. Work with your Owner team to establish timeframe expectations and regular progress updates.

4C. TOWN PLANNING & PERMITTING CONSULTANT: PHASE 1 & PHASE 2

SUMMARY:

The town planning and permitting consultant serves as a liaison between the primary design consultant firm and the Owner, to provide expert guidance on culture, climate regionally available materials as well as zoning, codes, ordinances and permitting process. As directed by the Owner, consultant facilitates any additional due diligence work on the property that may be required.

REQUIREMENTS: (sequential)

1. **INITIAL SITE INVESTIGATION:** Prepare a preliminary planning report including the following information:
 - Architectural and landscape plans, civil and storm water management plans
 - Relevant territorial authority(ies)
 - Planning statutory documents requiring consideration
 - Site zoning and any specific overlays or controls
 - Approval pathway considering the proposed land use and capital cost of the project
 - Potential site contamination
 - Notable or protected trees or vegetation, tree removal and replacement guidelines
 - Heritage or cultural sites impact assessment and other sensitive uses
 - Indigenous people planning documents
 - Archaeological assessment requirements
 - Hazard areas or flood plains on or near the site
 - Acoustic and visual impact assessments
 - Traffic assessment requirements and any potential access issues
 - Utility services report including any under-capacity existing services
 - Geotechnical restrictions and further assessments required
 - Detailed site survey
 - Demolition and waste management plans
2. **MEETINGS:** Lead necessary communications and meetings with the planning authority to determine key issues for planning proposal submission. Meet with the Owner and consultant design team regularly to update milestones and schedule, and to work through design considerations impacting permitting.
3. **ASSEMBLE DOCUMENTATION:** Determine all required documentation for Planning Permissions, and under the direction of the SPD Project Manager, gather existing reports/information already prepared and arrange for any additional reports necessary for Planning submission and approvals. Include all items included in initial site investigations as well as a detailed cost estimate and any other items required by the authority having jurisdiction.
4. **PLANNING SUBMISSIONS:** Prepare and or manage the preparation from other consultants, all necessary documentation, reports and information necessary for all required formal planning submissions to the local municipality.
5. **MUNICIPALITY COORDINATION:** As required throughout the planning preparation, submission and approval process, manage ongoing correspondence and follow-up communication with the local municipality until all planning approvals are obtained.

6. **PUBLIC NOTIFICATIONS:** If a public notification process is required, prepare all necessary planning submissions, required meetings with consultants, local municipality and others, prepared required evidence for hearings, attend all required meetings, hearing proceedings as required to obtain final planning approvals.
7. **PLANNING APPROVALS:** Manage all other necessary planning steps, submissions, follow-up correspondence with local municipality and all other steps required to obtain all the required planning approvals for the development.
8. **PLANNING APPROVAL SCHEDULE:** Establish a schedule for the overall planning preparation, submission and approval process. Identify key milestone dates for consultant engagement to include pre-lodgement requirements, completion of reports and planning documentation, formal planning submissions and estimated planning approvals.

SCHEDULE:

Town planning and permitting consultancy work may continue during both Phase 1 and Phase 2 and is considered complete when all required project planning approvals have been granted and the project is approved to move forward. Jurisdictional requirements for completion may vary around the world. Coordinate with the Owner and project team to determine completion.

4D. PEER REVIEW CONSULTANT: ARCHITECTURE & INTERIOR DESIGN

SUMMARY:

Peer review services provide a neutral professional review of documents created by the local design consultant, on behalf of the Owner Team. The Peer Review does not replace Owner reviews but provides an extra measure of review and coordination. Peer Reviewers are responsible to understand the “why’s” associated with the owner’s standards, review documents for conformance with program, guidelines, Owner requirements, and quality control coordination between architecture and subconsultant drawings typically in the Owner provided review platform. Additional mentoring or training in design and construction contract administration services may also be requested by the Owner, as needed. Liability for the correctness and accuracy of the drawings is not transferred to the Peer Reviewer in any way.

REQUIREMENTS:

1. **DRAWING REVIEW:** Conduct a thorough review of all design documents, including drawings, specifications, and reports, for consistency, coordination between disciplines, accuracy and Owner requirement conformance. Identify any corrections needed in the Owner review platform. Ensure that any revisions and updates are coordinated across all disciplines to maintain the integrity of the design. As directed by the Owner team, provide mentoring, review and feedback specific to Contract Administration services (RFI/submittals/deferred submittal (tendering) review).
2. **BUILDING ENVELOPE:** Review the design and construction of the building envelope, including exterior walls, windows, and doors. Ensure that these elements provide adequate thermal and moisture protection and are coordinated with the overall architectural and structural design. Ensure penetration details, foundation and wall systems show proper material systems, required overlaps and other technical best practices. Identify items needing correction in the Owner review platform.
3. **MENTORING & TRAINING:** As directed by the Owner provide mentoring and training to local consultant on program and guideline integration, scope of design work and detailing, scheduling and phasing of work and other items as identified. If requested by the Owner, assist the local consultant with training regarding the oversight of the construction process, RFI and submittal responsibilities and coordination. Serve as a resource and guide for the local consultant as they navigate the design and construction process. Provide progress updates to the Owner.
4. **PROGRAM AUDIT:** Perform a complete review of all facility program requirement with the Owner supplied Audit form. Document items that meet program, those that are altered to fit and those that are not included or incorrectly shown. The excel will be saved on the Owner management platform and all parties will work in the original document.
5. **QUALITY CONTROL:** Review architectural and engineering drawings for complete coordination, including verification of tags and labels, schedules, details and notes. Identify items needing correction in the Owner review platform.
6. **SPECIFICATIONS:** Review specification sections as requested by the Owner.

7. **LANDSCAPE:** Ensure feasibility studies and reports are provided for topography, geotechnical, environmental impact, and erosion control are provided and are in alignment with owner standards. Verify utility connections, structural and MEP infrastructure, site grading, stormwater management, landscaping, and roadways/pavement are per the owner's standards and provided localized solutions.

SCHEDULE

Peer Review Services must primarily be completed in conjunction with the Owner review period at the end of each deliverable as part of Phase 2 work. The review period is typically two weeks following submission by the local consultant. Comments will be provided on the Owner review platform.

4E. PEER REVIEW CONSULTANT: ENGINEERING

STRUCTURAL REQUIREMENTS:

1. **LOAD-BEARING ELEMENTS:** Ensure that load-bearing elements such as beams, columns, and walls are correctly aligned and coordinated with the overall structural design. This includes verifying that these elements can support the intended loads, stair movement from building and comply with design specifications and safety standards.
2. **FOUNDATION SYSTEMS:** Review the design and implementation of foundation systems to ensure they are suitable for the site conditions and can adequately support the building's structural loads. This includes checking for compatibility with geotechnical reports and ensuring proper integration with the overall structural design.
3. **STRUCTURAL CALCULATIONS:** Ensure that structural calculations are provided for seismic, wind dead and alive loads. This involves verifying that structural elements are designed to withstand these forces and that appropriate measures are in place to protect the building's integrity during such events.
4. **MODELS, PROGRAM FILES AND REPORTS:** Ensure that structural reports, modeling files, software programs, studies and calculation documents are complete and delivered to the owner.

MECHANICAL, ELECTRICAL & PLUMBING (MEP) REQUIREMENTS:

1. **HVAC SYSTEMS:** Verify the overall design meets the owner's guidelines and provides system comparison models and heat load calculations. Verify the overall design and layout of HVAC systems to ensure they meet the building's heating, cooling, ventilation, moisture level requirements and provide off-site communication according to the owner's guidelines. Verify that the designs incorporate locally sustainable/practiced, energy-efficient technologies are not over engineered and do not have unnecessary safety factors. Verify systems have received off-site and facility management communication according to the owner's guidelines. Ensure systems have correct corrosion protections for application. Ensure that these systems are integrated with the architectural and structural elements of the building to avoid conflicts and ensure efficient operation.
2. **ELECTRICAL SYSTEMS:** Verify the overall design meets the owner's electrical, communication, security, audio-visual guidelines and that defined roles between owner and contractor are identified for supplying equipment and installation. Ensure utility history reports are generated and are used when designing the distribution system. Verify that the designs incorporate locally sustainable/practices, energy-efficient technologies, are not over engineered and do not have unnecessary safety factors especially related to the electrical distribution sizing, distribution back up sizing, branch circuitry, lighting quantities, lighting levels, outlets and electrical loads. Ensure lighting levels in the Owner's standards and guidelines are achieved that include the decorative lighting, and renderings have been approved by the owner. Verify the systems have received off-site and facility management communication according to the Owner's standards and guidelines. Verify the proposed network integrated systems have been vetted by the owner's network security department. Ensure the intrusion and security are designed according to the site risk rating. Verify systems have correct corrosion protections for application.

Ensure that these systems are integrated with the architectural and structural elements of the building to avoid conflicts and ensure efficient operation.

3. **PLUMBING SYSTEMS:** Verify the overall design meets the Owner's standards and guidelines and that potable water has been analyzed and managed according to the Owner's standards and guidelines. Verify designs incorporate locally suitable, energy-efficient technologies, are not over engineered and do not have unnecessary safety factors. Ensure the font system is receiving the correct Net Positive Suction Head (actual/required) to avoid cavitation. Verify the systems have received off-site and facility management communication according to the Owner's standards and guidelines. Ensure systems have correct corrosion protections for application. Ensure that these systems are integrated with the architectural and structural elements of the building to avoid conflicts and ensure efficient operation.
4. **FIRE SUPPRESSION SYSTEMS:** Verify the overall design meets the Owner's standards and guidelines. Verify the placement and routing of fire suppression systems, including sprinklers and that the system is integrated with the fire alarm. Ensure systems have correct corrosion protections for application. Ensure that these systems are integrated with architectural and structural design and comply with fire safety codes and standards. Ensure that these systems are integrated with the architectural and structural elements of the building to avoid conflicts and ensure efficient operation.
5. **SYSTEM CONTROLS/MONITORING:** Ensure the overall design meets the Owner's standards and guidelines and that the Niagara platform is used for each controller. Ensure the Building Management System and or the Equipment Monitoring System identifies system architecture as outlined in the Owner's standards and guidelines and that the sequence of operation is clearly communicated. Ensure a commissioning plan is provided and that defined roles between owner and contractor are identified.
6. **MODELS, PROGRAM FILES AND REPORTS:** Ensure that utility reports, modeling files (HVAC energy/comparison, heat loading, lighting, electrical short circuit, electrical distribution coordination, PV (if applicable) acoustic, structural), software programs, studies and calculation documents are completed and delivered to the owner.

SCHEDULE

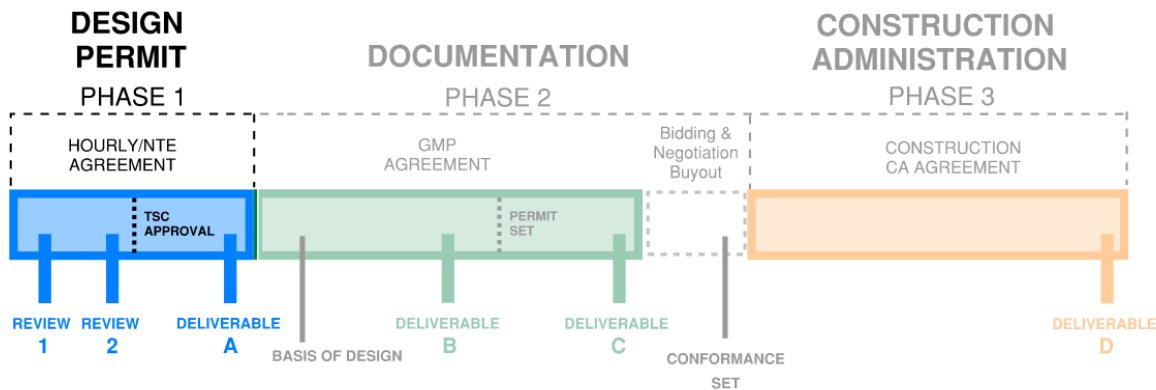
Peer Review Services must primarily be completed in conjunction with the Owner review period at the end of each deliverable as part of Phase 2 work. The review period is typically two weeks following submission by the local consultant. Comments will be provided on the Owner review platform.

4F. QUANTITY SURVEYING

PHASE 1

SUMMARY:

During Phase 1 the design team will work to develop conceptual site plan, floor plans, and exterior and interior renderings. The Quantity Surveyor participates minimally with the design team in Phase 1 to advise on cost impacts related to design decisions and directions.



REQUIREMENTS:

During Phase 1 the design team will work to develop conceptual site plan, floor plans, and exterior and interior renderings. The Quantity Surveyor participates minimally with the design team in Phase 1 to advise on cost impacts related to design decisions and directions.

1. **BUDGET:** The project team is responsible for providing design within the Owner's established project budget. Coordinate with the team and Owner in preparing, reviewing and maintaining design direction, design choices, and cost models that meet Owner's budget. Use the Owner provided standard construction cost estimate format.
2. **COST ESTIMATE:** Prepare a detailed cost estimate according to the schedule outline below after receipt of Deliverable A. Estimate to be based on approximate quantities of major items at current process and supplemented by unit costs derived from recorded cost data and analysis for items which cannot be quantified at each stage of planning.
3. **MEETINGS:** Attend periodic coordination meetings with the project team early to provide guidance and feedback on conceptual design impacts, material availability and cost implications.

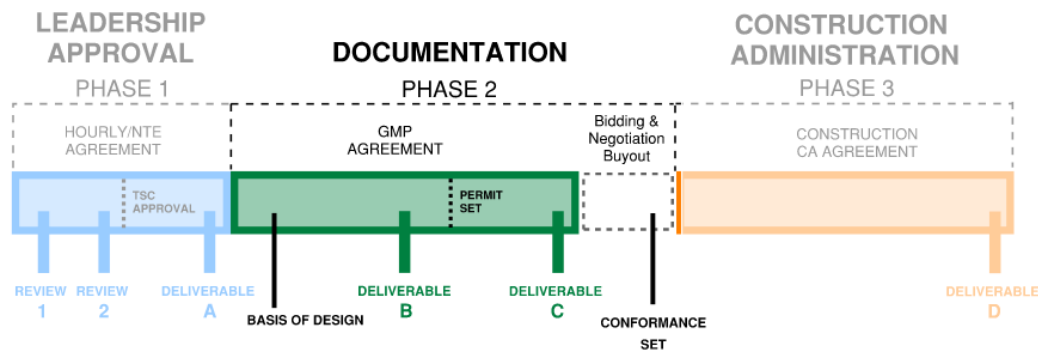
SCHEDULE:

Assist the Owner in establishing a construction cost budget for the project within 10 calendar days of the receipt of Deliverable A documents.

4G. QUANTITY SURVEYING PHASE 2

SUMMARY:

During Phase 2 the design team will develop and complete the construction documents. The Quantity Surveyor will prepare cost estimates for the Owner review, based on drawings and specifications at both Deliverable B (50 % project completion) and Deliverable C (100% project completion).



REQUIREMENTS:

1. **BUDGET:** The project team is responsible for providing design within the Owner's established project budget. Coordinate with the team and Owner in preparing, reviewing and maintaining design direction, design choices, and cost models that meet Owner's budget. Use the Owner provided standard construction cost estimate format. Participate in value engineering exercise in conjunction with the design team and owner of alternate design solutions, materials, etc. Update the cost plan progressively as the design develops and reconcile with the original construction cost budget.
2. **MEETINGS:** Attend and actively participate in design coordination meetings, as directed by the SPD Project Manager. Counsel with the design team about local/regional availability of all materials, systems and finishes. As invited by the SPD Project Manager, participate in preconstruction meetings.
3. **COST ESTIMATES:** Estimate all buildings and site elements, itemized by scope, to validate that design direction adheres to Owner budget requirements. Provide updated cost estimates according to the schedule outlined below, after receipt of Deliverable B and Deliverable C documents.
4. **COST COMPARISONS:** Obtain the General Contractor's cost estimate in the same Owner-provided format and compare the two estimates to highlight differences and to assist in reconciliation efforts.
5. **SITE VISIT:** At the request of the Owner's Project Manager, visit the project site to investigate and determine site conditions, construction means and methods employed in the region of the project, customs and culture in the area. Research, investigate and evaluate design and construction companies in the region as requested by the Owner.

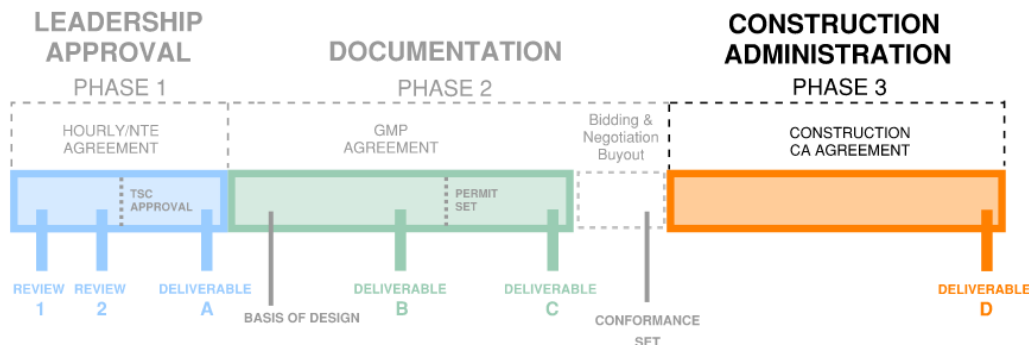
SCHEDULE:

An updated cost estimate will be provided within twenty-one calendar days of receipt of Deliverable B drawings, representing 50% project completion. An updated cost estimate will be provided within twenty-one calendar days of receipt of Deliverable C drawings, representing 100% project completion.

4H. QUANTITY SURVEYING: PHASE 3

SUMMARY:

Phase 3 includes the construction, commissioning and close-out of the project. Participate in construction contract administration services and oversight for the duration of the project. The project is not considered complete until the final completion report is received by the Owner.



REQUIREMENTS: (sequential)

1. **MEETINGS:** Attend and actively participate in OAC meetings and other project meetings, as directed by the SPD Project Manager.
2. **MILESTONE COST ESTIMATES:** Estimate all buildings and site elements, itemized by scope, to validate that design direction adheres to Owner budget requirements. Provide updated cost estimates within twenty-one calendar days of receipt of Deliverable B and Deliverable C documents.
3. **COST COMPARISONS:** Obtain the General Contractor's cost estimate in the same Owner-provided format and compare the two estimates to highlight differences and to assist in reconciliation efforts.
4. **PROCUREMENT/TENDERING:** Review the builder's take off and bid package summary, as prepared by the Contractor and the 100% Tendering Documents, and advise Owner and Contractor of any corrections, adjustments or modifications necessary to obtain competitive subcontractor and supplier bids for the project. Through the prequalification process, review the potential subcontractors and suppliers proposed by the Contractor, assist the Owner in verifying they can perform the work required and can meet Owner quality standards.
5. **BID REVIEW:** Assist Owner in reviewing Contractor's subcontractor and supplier bids, bid tabulations and recommendations for selection. Contractor will have multiple bid packages to include demolition, structure, façade, interiors, MEP, external and landscape at a minimum. Multiple subcontractor trades will be included in each package. Identify errors in arithmetic and inconsistent pricing, scrutinize the pricing strategy of bidders, compare bids to project budget, and advise the Owner of any action required.
6. **BID REPORT:** Provide a written report for the Owner with recommendations for any amendments to the bids and to the subcontract documents considered advisable, before the award of the subcontracts

for the project. Maintain a list of all modifications that will be incorporated into construction documents.

7. **GMP COST REVIEW:** Assist the Owner in analysis of Contractor's Cost Plus GMP to clarify the terms and pricing of the contract sum as defined in the contract documents, and in drafting a letter of acceptance.
8. **BUDGET:** Actively monitor construction budgets and notify Owner of risks or overages. Track the project with the relevant cost codes established together with the Owner Project Manager. Evaluate the current cost of these codes and the amount spent to-date for each cost code, incorporate into a monthly project financial standing report to Owner.
9. **SITE VISITS:** Visit the project site at SPD Project Manager's direction and as specified by contract and evaluate work completed to date for variations and payment applications, work with contractor to resolve discrepancies or disputes.
10. **PAYMENT APPLICATIONS:** Thoroughly review and evaluate each payment application submitted by Contractor. Compare work claimed to date to Bill of Quantities. Work with Contractor to resolve any discrepancies and forward a signed payment application to Owner within seven calendar days of receipt of payment application. In conjunction with the design consultant, prepare monthly valuations of outstanding deficiencies and recommend adjustments to the monthly payments to withhold the deficiency valuations until the deficiencies are remediated and confirmed complete by Owner.
11. **CHANGE ORDERS:** Assist Owner and design consultant in determining whether Contractor is entitled under the contract documents to a change in the scope of work that will affect cost or time, for completion of the work. If Contractor is entitled to a change order, provide to Owner and design consultant an estimate of the cost for the change in the scope. Review the breakdown of quantities, material unit prices, labor unit prices, overhead and profit to perform the change in the work. Review Contractor's price proposal and advise Owner regarding the reasonableness with reference to the bill of quantities and the overhead and profit permitted by the contract documents. Assist the Owner with negotiations with Contractor to establish the cost of the proposed change order and any related time adjustment.

If Owner and Contractor agree on the cost of the proposed change order and any related time adjustment, Owner will instruct Contractor to prepare and sign the proposed change order on an Owner approved form and send to Owner and design consultant for approval.

If Owner and Contractor do not agree on the cost of the proposed change order or any related time adjustment, Quantity Surveyor to prepare the form with the reasonable costs of the proposed change order and any related time adjustment and send it to the Owner and design consultant for final approval. (Construction Change Directive) Within ten calendar days of receiving notice of any controversies, claims or disputes related to the project whether raised by Owner or others, assess and report in writing to Owner on contractual and financial implications and assist Owner in negotiating such controversies, claims or disputes.

12. **FINAL COMPLETION:** When the work is practically or substantially complete, Quantity Surveyor will prepare a draft final account based on the contract documents, including as-built drawings, will confirm quantities and rates with Contractor to ensure accuracy, will identify discrepancies, and will provide reasons for proposed reconciliation of discrepancies. When the work has finally been completed and

Owner has confirmed acceptance, prepare the final account and present it to the Owner within sixty calendar days of the date of final completion.

13. **SERVICE EXCLUSIONS:** Project management and construction services, full-time project site-based staffing, services related to arbitration or litigation proceedings, owner purchased items, artwork, assessment of quality during progress payments, utility connection fees, permit fees.

SCHEDULE:

Procurement and Tendering services will be completed within eight months after completing Phase 2 services. Construction Administration services will be provided from Notice to Proceed until project completion. Final Completion services will be completed within sixty calendar days of the date of final completion.